Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, C/C++, SQL, R
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Multithreading & Concurrency
Software & Data Engineering: Data Processing, Data Pipelines, Database Design, REST APIs, Unit Testing, Test-Driven Development
Data Science & Analytics: Pandas, NumPy, Scikit-learn, PyTorch, Statistical Analysis, Feature Engineering, Time-Series Analysis, Anomaly Detection, Model Evaluation
Databases & Tools: MySQL, SQLite, Redis, Git, GitHub, Docker, Jupyter Notebook
Analytics & Visualization: Power BI, DAX, Tableau, Matplotlib, Exploratory Data Analysis

EXPERIENCE

VOIS for Tech University Engagement Program* |LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

***RedisLite* |LINK**

- Built a Redis-inspired in-memory data store in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, and O(1) LRU caching.
- Implemented TTL expiration, write-ahead logging, and crash recovery across a four-layer architecture.
- Added 46 automated PyTest unit tests and benchmarked 619K SET operations/sec in-memory and 2.8K operations/sec with durable WAL persistence.

***Energy Anomaly Detector*| LINK**

- Developed a Python pipeline to process electricity-consumption time series using hourly resampling, temporal feature engineering, rolling baselines, and Isolation Forest.
- Automated anomaly logging, visual reporting, and email alerts, with rate limiting to control alert frequency.
- Evaluated detections against a 95th-percentile threshold, identifying 9 anomalies across 6,918 test hours.

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Built an end-to-end pipeline for 50K+ PaySim transactions, incorporating duplicate detection, statistical outlier analysis, vendor-velocity checks, concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite warehouse and composite 0–100 risk score, identifying 357 high- or critical-risk transactions.
- Developed a five-page Power BI dashboard with 20+ DAX measures and automated pipeline tests; completed analysis in under 7 seconds.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** — 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** — 2025
- AWS APAC Solutions Architecture virtual experience program | **LINK** — 2025